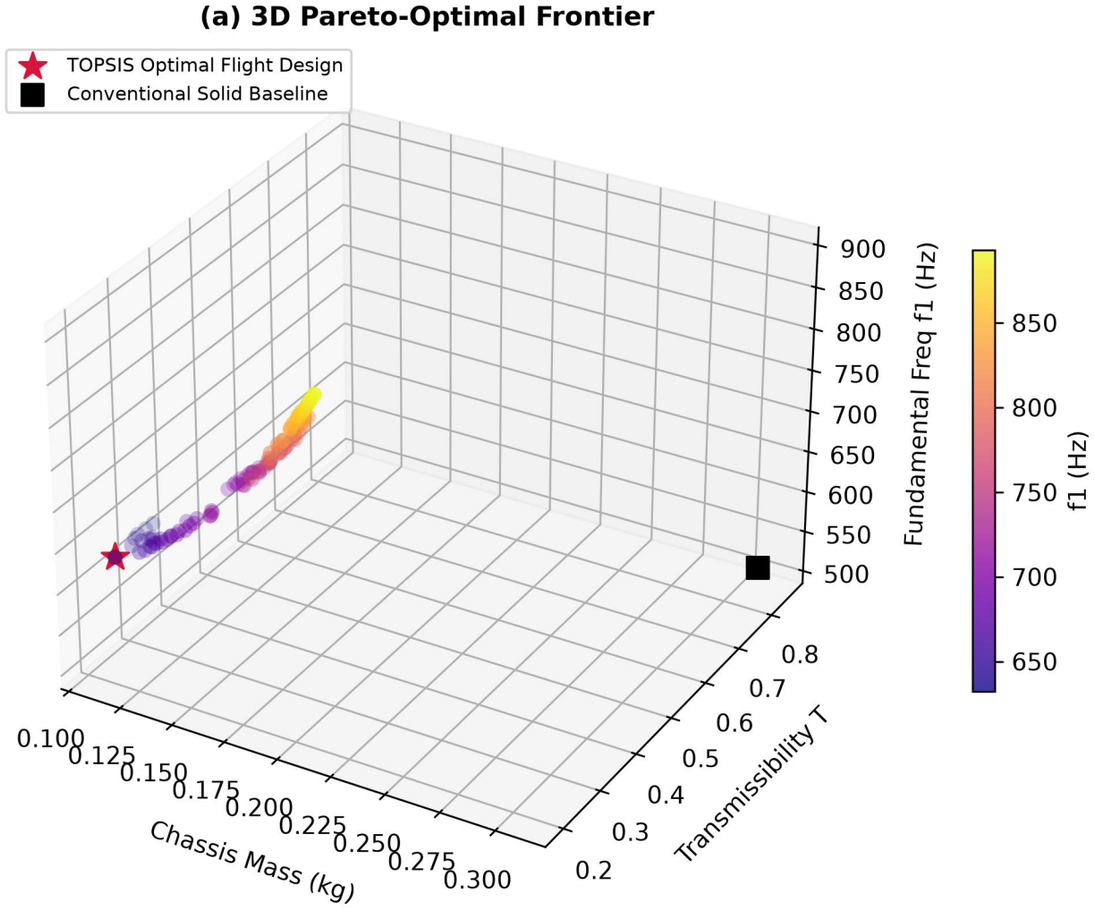
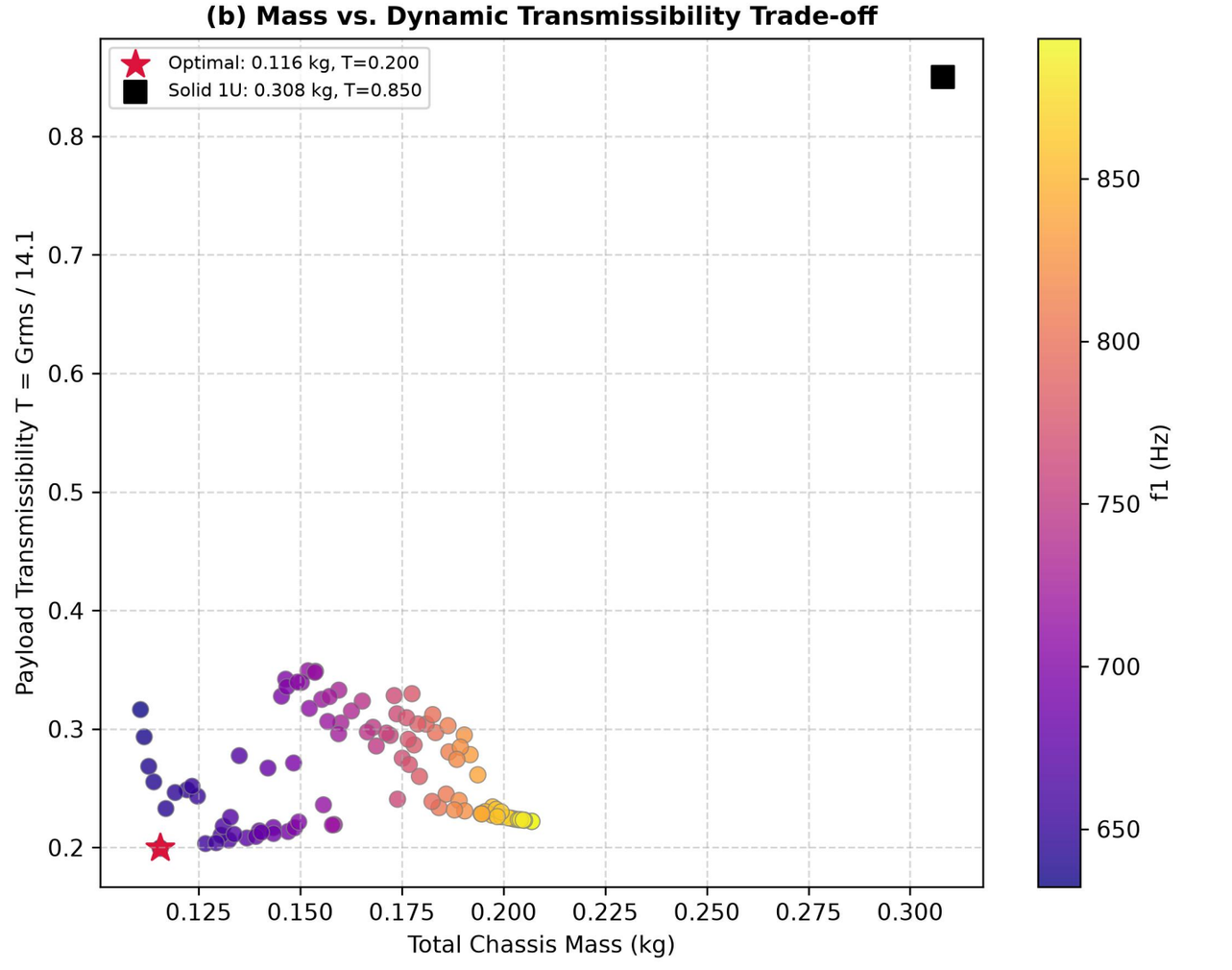
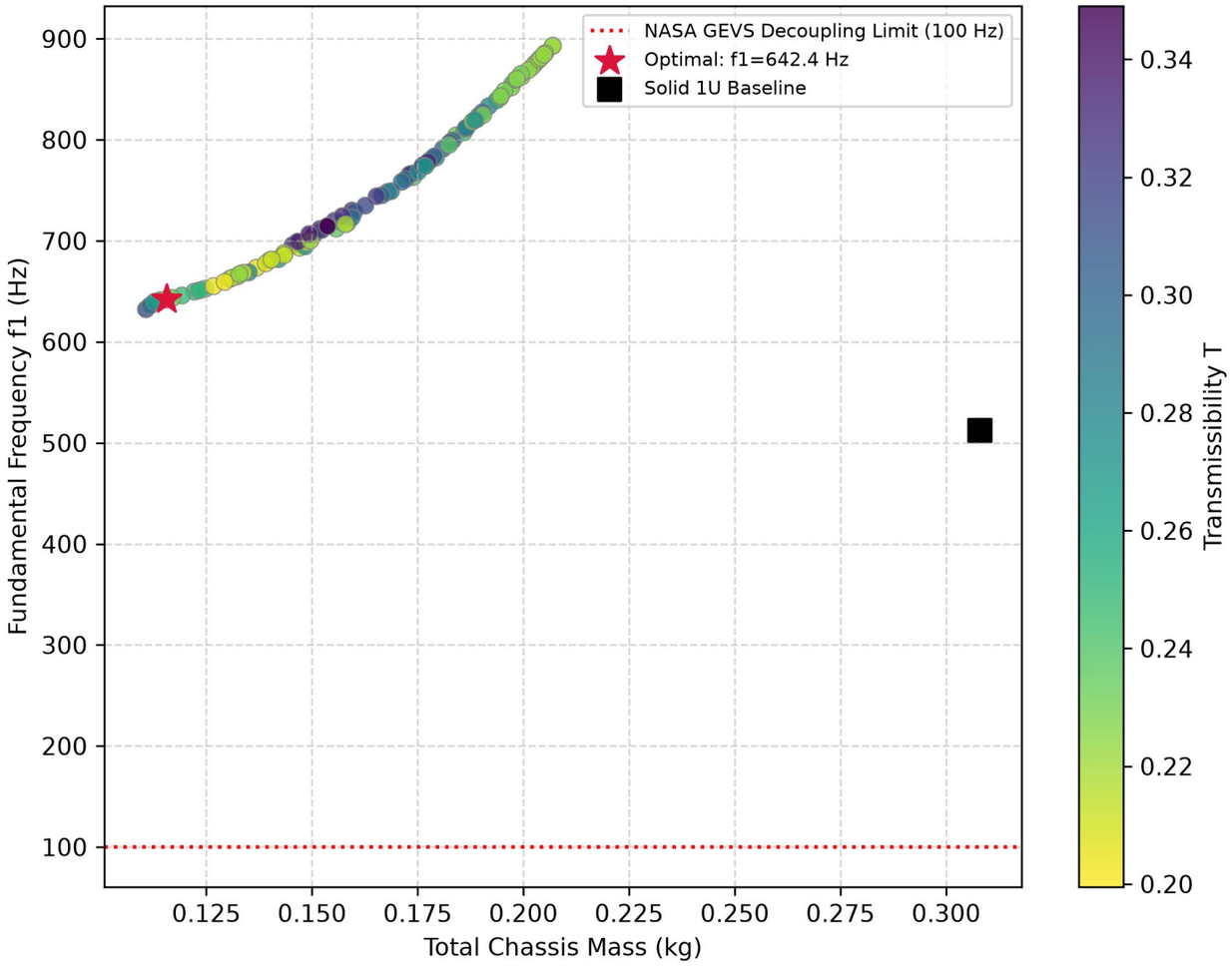
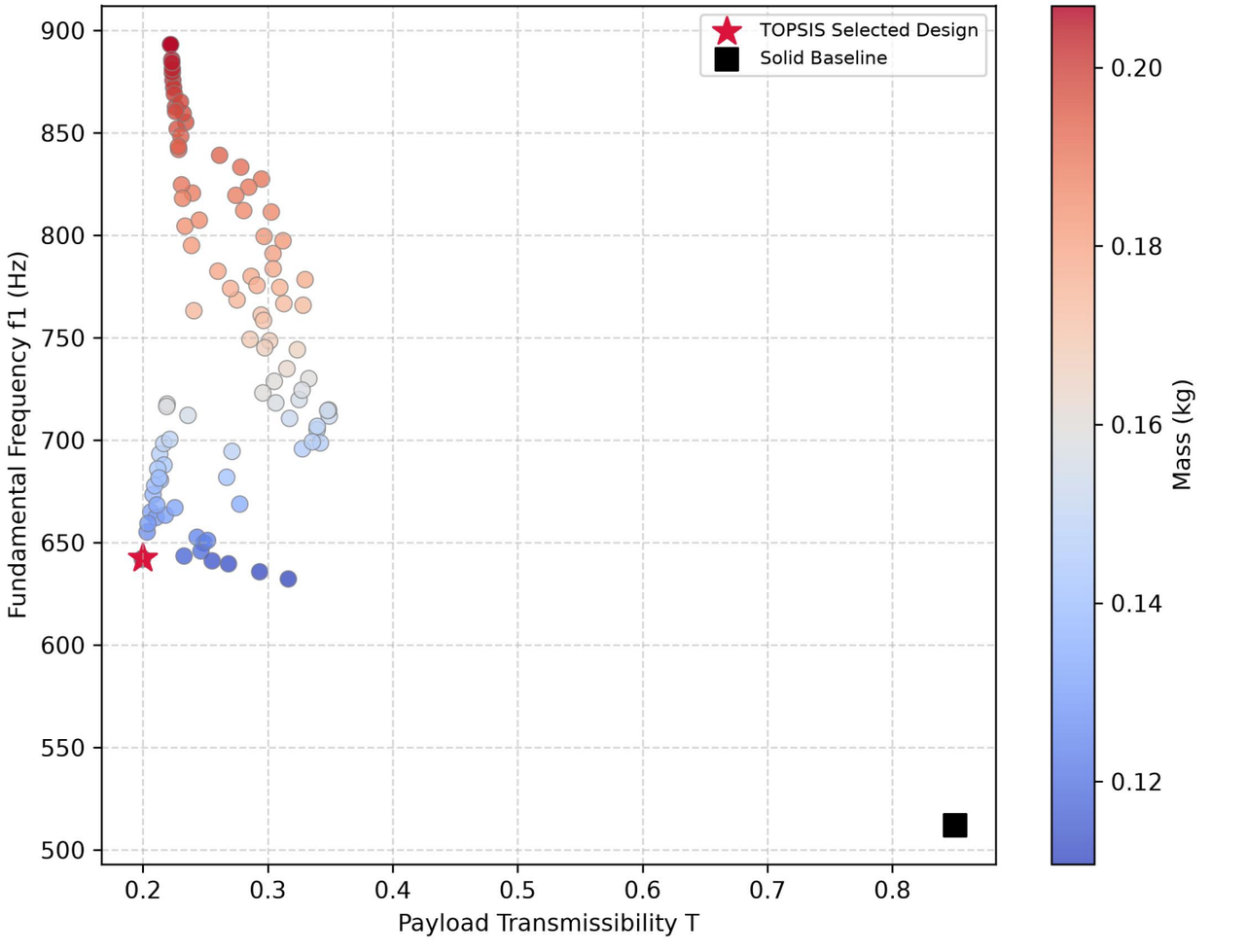

(a) 3D Pareto-Optimal Frontier
TOPSIS Optimal Flight Design
Conventional Solid Baseline
Chassis Mass (kg)
Transmissibility T
Fundamental Freq f1 (Hz)
f1 (Hz)


(b) Mass vs. Dynamic Transmissibility Trade-off
Optimal: 0.116 kg, T=0.200
Solid 1U: 0.308 kg, T=0.850
Payload Transmissibility T = Grms / 14.1
Total Chassis Mass (kg)
f1 (Hz)


(c) Structural Rigidity vs. Mass
NASA GEVS Decoupling Limit (100 Hz)
Optimal: f1=642.4 Hz
Solid 1U Baseline
Fundamental Frequency f1 (Hz)
Total Chassis Mass (kg)
Transmissibility T


(d) Dynamic Attenuation vs. Rigidity
TOPSIS Selected Design
Solid Baseline
Fundamental Frequency f1 (Hz)
Payload Transmissibility T
Mass (kg)